

Password Strength Analyzer with Custom Wordlist Generator

Internship Project Report | Elevate Labs | May 2026

1. Introduction

In the modern digital era, weak passwords remain one of the leading causes of security breaches and unauthorized access. Users often choose passwords based on personal information such as names, dates of birth, or pet names, making them highly susceptible to targeted attacks. This project aims to address this problem by building a Python-based tool that evaluates password strength and generates custom attack-specific wordlists based on user-provided personal data. The tool serves both as an educational resource and a practical utility for cybersecurity awareness.

2. Abstract

This project presents a Password Strength Analyzer with a Custom Wordlist Generator developed using Python. The tool accepts a password as input and evaluates its strength using the `zxcvbn` library, providing a score from 0 to 4, an estimated crack time, and improvement suggestions. Additionally, the tool accepts personal inputs such as name, date of birth, and pet name to generate a customized wordlist with variations including leetspeak substitutions, year suffixes, and capitalization patterns. The wordlist is exported as a `.txt` file. A graphical user interface (GUI) was built using Tkinter to make the tool accessible and user-friendly.

3. Tools Used

- **Python 3.x** — Core programming language used for the entire project.
 - **zxcvbn** — Password strength estimation library that provides score, crack time, and feedback.
 - **argparse** — Standard Python library used to build the command-line interface (CLI).
 - **tkinter / ttk** — Python's built-in GUI library used to build the graphical interface.
 - **VS Code** — Code editor used to write, run, and debug the project.
-

4. Steps Involved in Building the Project

Step 1: Environment Setup

Installed Python 3.x and the `zxcvbn` library using `pip`. Created a project folder and set up VS Code as the development environment.

Step 2: Password Analyzer Module

Used the `zxcvbn` library to analyze password strength. The module accepts a password input and returns a score (0–4), estimated crack time, and suggestions for improvement.

Step 3: Wordlist Generator Module

Built a function to accept personal inputs (name, date of birth, pet name) and generate password variants using leetspeak substitutions (e.g., `a→@`, `e→3`), year suffixes (2023, 2024, 2025), and capitalization patterns. The output is saved as a `.txt` file.

Step 4: CLI Interface

Used argparse to create a clean command-line interface with two modes: --analyze for password evaluation and --generate for wordlist creation.

Step 5: GUI with Tkinter

Developed a graphical user interface using Tkinter featuring a password input field, strength progress bar with color coding, crack time display, tips section, and a wordlist generator panel with export button. Added mouse-wheel scrolling support for full content visibility.

Step 6: Testing

Tested the tool with weak, medium, and strong passwords to verify accuracy of scores. Tested the wordlist generator with various personal inputs to validate output quality.

5. Conclusion

The Password Strength Analyzer with Custom Wordlist Generator was successfully developed and tested. The project demonstrates a practical application of Python in the cybersecurity domain, covering password analysis, pattern-based wordlist generation, CLI design, and GUI development. The tool effectively highlights the risk of using personal information in passwords and educates users about stronger password practices. This project provided hands-on experience with Python libraries, GUI development, and core cybersecurity concepts such as entropy, crack time estimation, and attack wordlists.